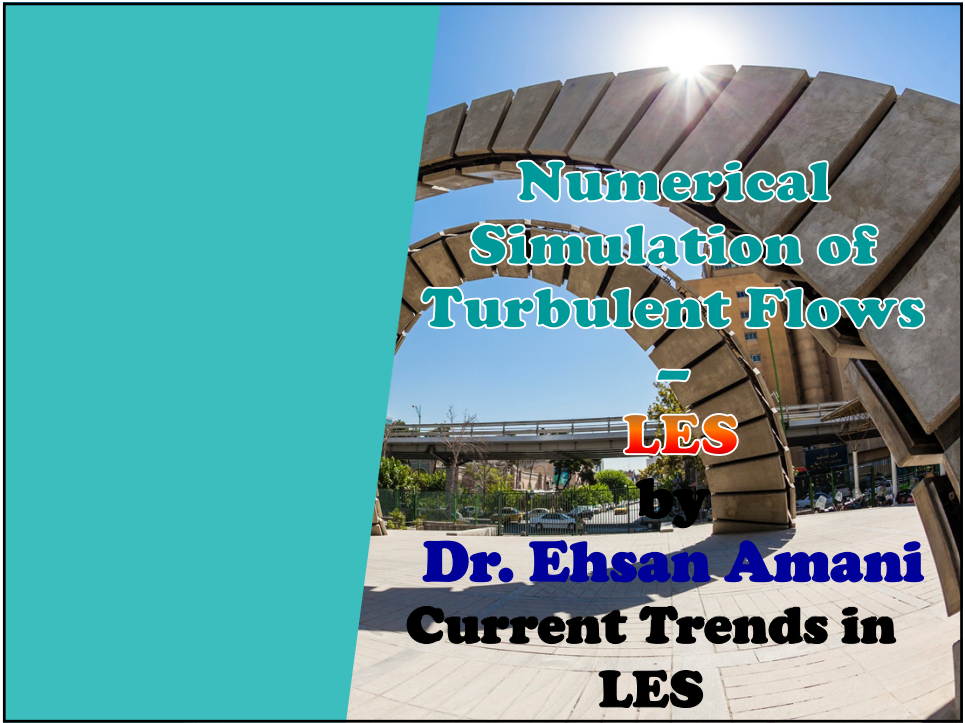
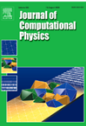




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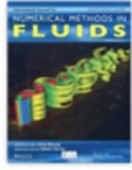
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2

Implicit LES (ILES)

Volpiani, Pedro Stefanin. "A comprehensive study about implicit/explicit large-eddy simulations with implicit/explicit filtering." *Flow, Turbulence and Combustion* (2024): 1-32.

- What is ILES?
- Relaxation filtering vs. Convolution filters

$$u^f(x_0) = u(x_0) - \chi D(x_0), \text{ with } D(x_0) = \sum_{j=-N_p}^{N_p} d_j u(x_0 + j\Delta x)$$

- Solution stabilization: split convective flux

- Feiereisen *et al.* [28]'s formulation applied only to the momentum and energy equations:

$$C = \frac{1}{2}C^D + \frac{1}{2}C^\phi$$

- Ducros *et al.* [30]'s formulation for the mass, momentum and energy equations:

$$C = \frac{1}{2}C^D + \frac{1}{2}C^u$$

- and Kennedy and Gruber [31]'s formulation for the mass, momentum and energy equations:

$$C = \frac{1}{4}(C^D + C^u + C^\phi + C^p)$$

$$\begin{aligned} C^D &= \frac{\partial \rho u_j \phi}{\partial x_j}, \\ C^\phi &= \phi \frac{\partial \rho u_j}{\partial x_j} + \rho u_j \frac{\partial \phi}{\partial x_j}, \\ C^u &= u_j \frac{\partial \rho \phi}{\partial x_j} + \rho \phi \frac{\partial u_j}{\partial x_j}, \\ C^p &= \rho \frac{\partial u_j \phi}{\partial x_j} + u_j \phi \frac{\partial \rho}{\partial x_j}. \end{aligned}$$

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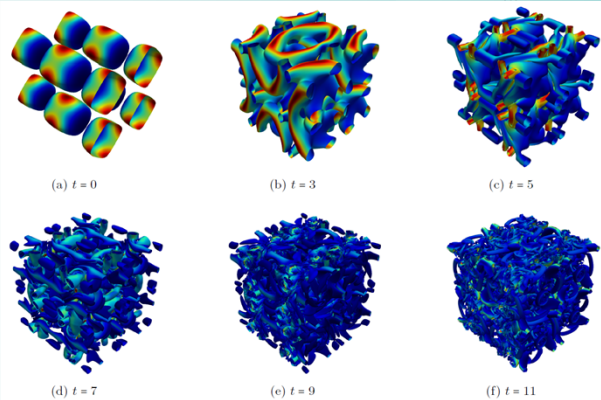
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Implicit LES (ILES)

Volpiani, Pedro Stefanin. "A comprehensive study about implicit/explicit large-eddy simulations with implicit/explicit filtering." *Flow, Turbulence and Combustion*

- Benchmark: Taylor-Green Vortex

$$\begin{aligned} u_0(\mathbf{x}) &= U_\infty \sin(x) \cos(y) \cos(z) \\ v_0(\mathbf{x}) &= -U_\infty \cos(x) \sin(y) \cos(z) \\ w_0(\mathbf{x}) &= 0 \\ p_0(\mathbf{x}) &= p_\infty + \frac{q_\infty}{16} [\cos(2z) + 2] [\cos(2x) + \cos(2y)] \end{aligned}$$



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Wall-modeled LES (WMLES)

Fowler, Mitchell, Tamer A. Zaki, and Charles Meneveau. "A multi-time-scale wall model for large-eddy simulations and applications to non-equilibrium channel flows." *Journal of Fluid Mechanics* 974 (2023): A51.

- Wall-function vs. Wall-stress model

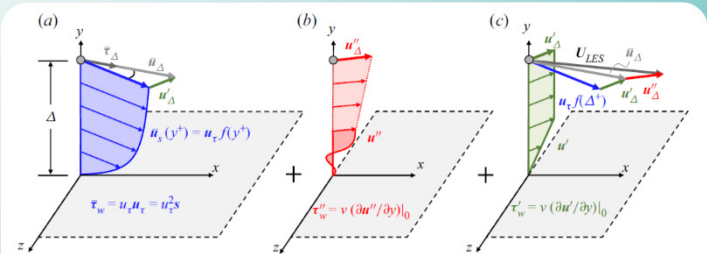


Figure 2. A schematic of the velocity profiles for the (a) LaRTE, (b) laminar non-equilibrium, and (c) turbulent non-equilibrium components. Additional visualizations (including an interactive animation) of the various terms comprising the MTS wall model are provided in Appendix A, in figure 27.

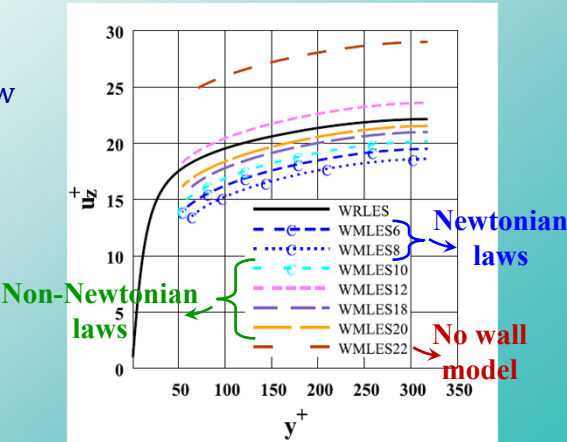
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Wall-modeled LES (WMLES)

Taghvaei, Mohammad, and Ehsan Amani. "Wall-modeled large-eddy simulation of turbulent non-Newtonian power-law fluid flows." *Journal of Non-Newtonian Fluid Mechanics* 322 (2023): 105136.

- Non-Newtonian flow additional closure



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Advanced SGS closures

Amani, Ehsan, Mohammad Bagher Molaei, and Morteza Ghorbani. "Novel mixed approximate deconvolution subgrid-scale models for large-eddy simulation." *Physics of Fluids* 36.8 (2024).

- Explicit filtering

$$\begin{aligned} \partial_i \bar{\bar{u}}_i + \partial_j (\bar{\bar{u}}_i \bar{\bar{u}}_j) &= -\partial_i \bar{p} + \partial_j (2\nu \bar{\bar{S}}_{ij}) - \partial_j T_{ij}, \\ \partial_i \bar{\bar{u}}_i &= 0, \end{aligned} \quad T_{ij} \equiv \overline{u_i u_j} - \bar{\bar{u}}_i \bar{\bar{u}}_j = b_{ij} + a_{ij}.$$

- Deconvolution: Van-Cittert iteration

$$\begin{aligned} u_0^* &= \bar{\bar{u}}, \\ u_n^* &= u_{n-1}^* + \left(\bar{\bar{u}} - G * u_{n-1}^* \right); \quad n = 1, 2, \dots, N, \end{aligned}$$

- Hard- vs. soft- deconvolution
- Consistent SGS closures

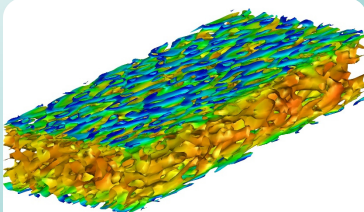
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Advanced SGS closures

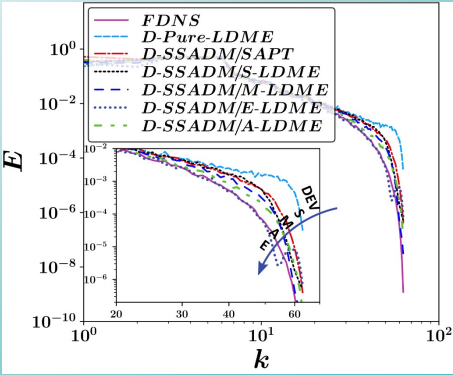
Amani, Ehsan, Mohammad Bagher Molaei, and Morteza Ghorbani. "Novel mixed approximate deconvolution subgrid-scale models for large-eddy simulation." *Physics of Fluids* 36.8 (2024).

- Benchmark: 1D Burgers turbulence
- Benchmark: Turbulent channel flow



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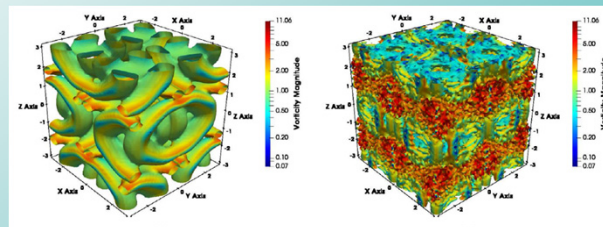
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Explicit filter design

Najafiyazdi, Mostafa, Luc Mongeau, and Siva Nadarajah. "Large eddy simulation on unstructured grids using explicit differential filtering: A case study of Taylor-Green vortex." *Journal of Computational Physics* 476 (2023): 111833.

- Explicit vs. implicit filtering
- Ideal filter properties

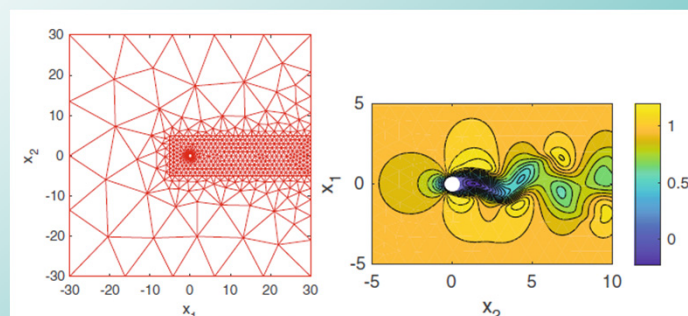


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Discontinuous Galerkin

Anantharamu, Sreevatsa, and Krishnan Mahesh. "A Non-Dissipative, Energy-Conserving, Arbitrary High-Order Numerical Method and Its Efficient Implementation for Incompressible Flow Simulation in Complex Geometries." *International Journal for Numerical Methods in Fluids* 97.4 (2025): 503-522.



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